

AI-BASED COUGH ANALYSIS USING RASPBERRY PI

Formal Project Idea Report

1. Project Overview

This project proposes a low-cost, edge-AI healthcare screening system that records cough sounds using a microphone connected to a Raspberry Pi and analyzes the audio using a machine-learning model. The system extracts meaningful acoustic features from cough recordings and classifies them into predefined cough or respiratory-sound categories. The analysis is performed locally on the Raspberry Pi, reducing the need to continuously transmit sensitive audio data to a cloud server.

2. Problem Statement

Cough is a common symptom associated with a wide range of respiratory conditions, but its acoustic characteristics can vary considerably. Conventional assessment may require clinical examination and specialized equipment. The proposed project investigates whether machine learning can identify useful patterns in cough sounds using an inexpensive embedded computing platform. The system is intended as a screening and research prototype, not as a replacement for professional medical diagnosis.

3. Objectives

- Develop a Raspberry Pi-based system for recording cough sounds.
- Preprocess cough audio and extract relevant acoustic features.
- Train and evaluate machine-learning/deep-learning models for cough classification.
- Deploy a lightweight model capable of local inference on Raspberry Pi.
- Measure classification performance as well as inference time and resource usage.
- Investigate whether model optimization can improve edge-device performance while retaining useful accuracy.

4. Proposed System

Audio Capture → Preprocessing → Feature Extraction → ML Model → Classification → Result

A microphone captures a cough recording. The audio is cleaned and segmented, followed by conversion into a suitable representation such as a Mel spectrogram or extraction of MFCC features. A trained classification model then analyzes the representation and produces a predicted cough category or respiratory-abnormality score. The result can be displayed through a simple web dashboard or local interface.

5. Artificial Intelligence Methodology

Stage	Proposed Approach
-------	-------------------

Audio preprocessing	Noise reduction, normalization, segmentation
Feature representation	MFCCs and/or Mel spectrograms
Candidate models	CNN, MobileNet, SVM, Random Forest
Optimization	Quantization and lightweight model deployment
Output	Cough category / respiratory-pattern screening score

6. Hardware Requirements

- Raspberry Pi 4 or Raspberry Pi 5
- USB microphone or suitable digital microphone
- MicroSD card and power supply
- Optional: small display, LEDs, or speaker for local feedback
- Optional future extension: pulse oximeter, temperature sensor, or other physiological sensors

7. Software and Dataset

The prototype can be implemented using Python, NumPy, Librosa, OpenCV where required, scikit-learn, TensorFlow/PyTorch, and an edge deployment framework such as TensorFlow Lite or ONNX Runtime. Publicly available respiratory-audio datasets, such as COUGHVID or Coswara, can be investigated for model training and evaluation, subject to their licensing and usage conditions. A small locally collected dataset may also be used for controlled testing with appropriate consent and privacy safeguards.

8. Research Component

The project can be extended into a research study by comparing different models and measuring the trade-off between predictive performance and edge-device efficiency. Important experimental measures include accuracy, precision, recall, F1-score, confusion matrix, inference latency, RAM usage, CPU utilization, and power consumption. A particularly relevant research question is whether a quantized lightweight model can provide acceptable cough-classification performance while substantially reducing inference cost on a Raspberry Pi.

9. Expected Outcome

The expected outcome is a functional Raspberry Pi prototype capable of recording cough sounds and performing local AI inference. The project should demonstrate that a compact machine-learning model can process audio in near real time on an affordable edge device. The final system should provide a clear screening-oriented output rather than claiming definitive diagnosis.

10. Innovation and Significance

The main value of the project is the combination of healthcare audio analysis, machine learning, and edge computing. Unlike a cloud-only solution, local inference can improve privacy and reduce dependence on network connectivity. The research component can further investigate multimodal extension by combining cough audio with physiological measurements such as SpO₂ or heart rate.

11. Limitations and Ethical Considerations

Cough sounds can be influenced by recording conditions, microphones, background noise, age, language, and individual differences. Dataset bias may therefore affect model performance. The system must not be presented as a medical diagnostic device without appropriate clinical validation and regulatory approval. Any newly collected recordings should be obtained with informed consent, securely stored, and de-identified where possible.

12. Future Scope

- Multimodal analysis using cough audio, SpO₂, heart rate, and temperature.
- Personalized models that account for individual baseline cough characteristics.
- Continuous or on-demand monitoring through a portable enclosure.
- Model compression and hardware acceleration for faster edge inference.
- Clinical validation using appropriately collected and annotated datasets.

13. Proposed Project Title

“Edge AI-Based Cough Sound Analysis and Respiratory Screening Using Raspberry Pi”

Note: This document describes a project concept for academic and research purposes. It is not a medical diagnostic protocol and should not be used to make clinical decisions.